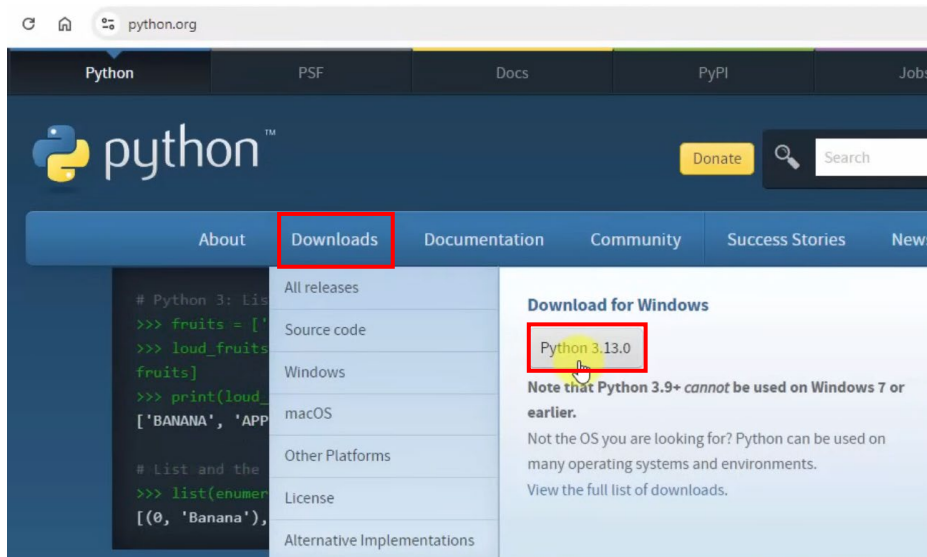


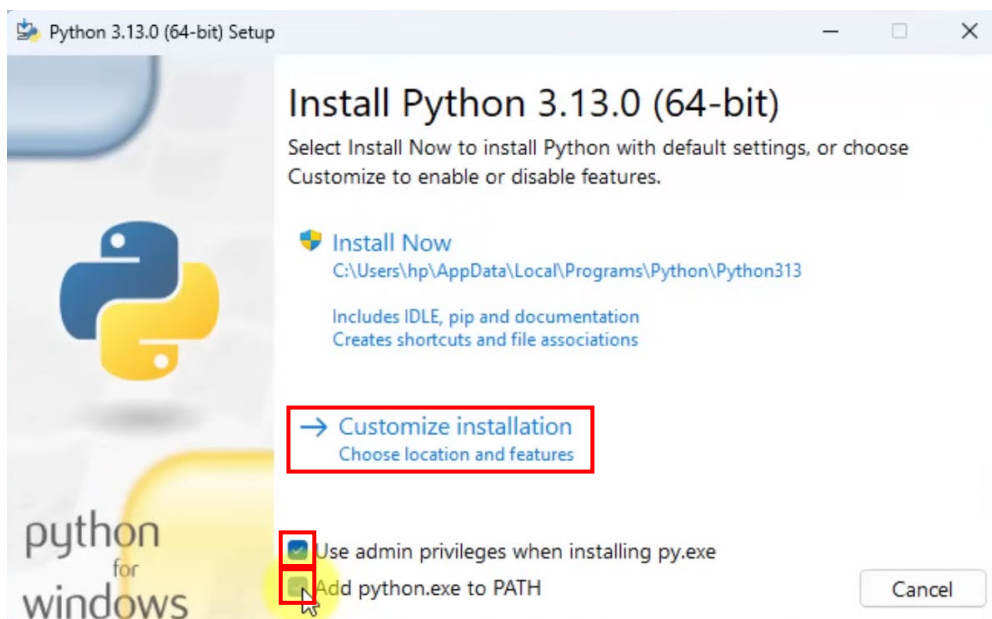
Install Python environment on your Windows

1. Click Downloads > Python 3.x.x

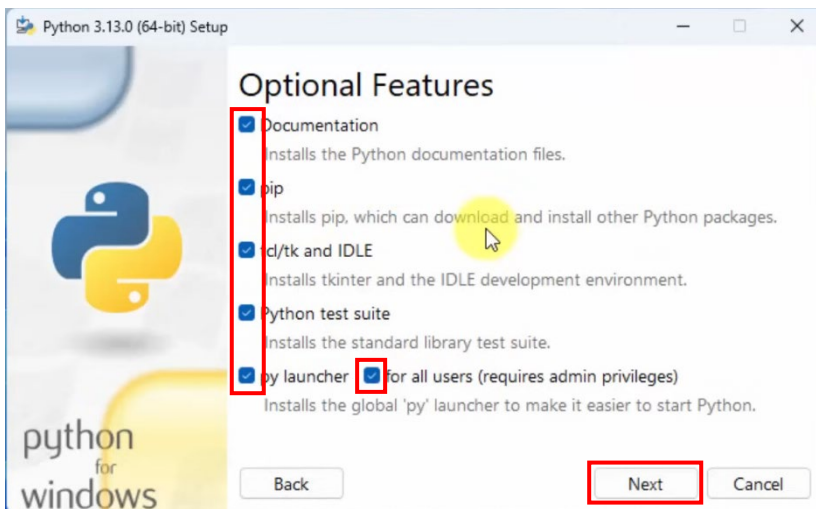


2. Check:

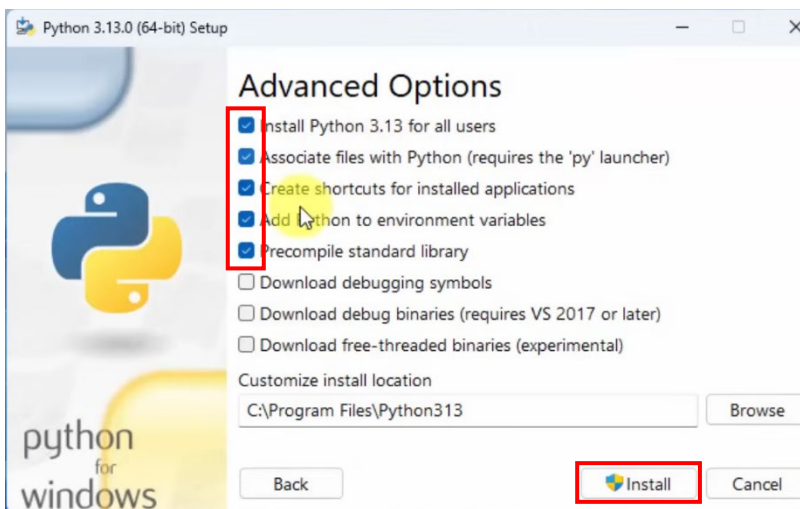
- Use admin privileges when installing py.exe
 - Add python.exe to PATH
- > Customize installation



3. Check all > Next



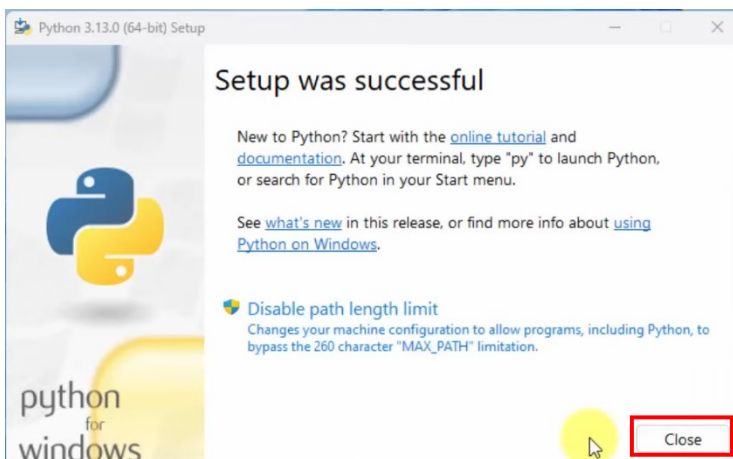
4. Advanced options



Check:

- Install Python 3.13 for all users
- Associate files with Python
- Create shortcuts for installed applications
- Add Python to environment variables
- Precompile standard library
- > Install

5. Complete installation: click Close



ODBC SQL Server Connection

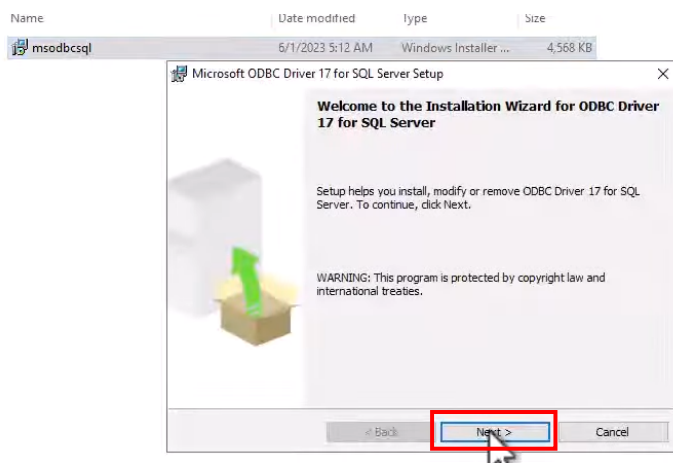
6. Install ODBC Driver17

Url: <https://learn.microsoft.com/en-us/sql/connect/odbc/download-odbc-driver-for-sql-server?view=sql-server-ver17>

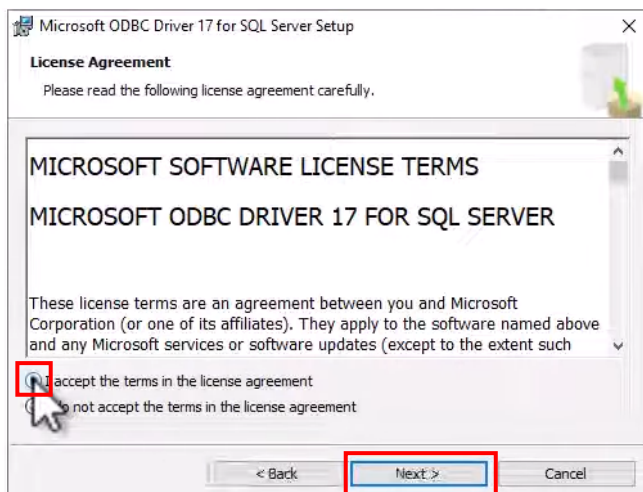
The screenshot shows the Microsoft ODBC Driver 17 for SQL Server download page. The page title is "Microsoft ODBC Driver 18.6.2.1 for SQL Server (ARM64): Chinese (Simplified) | Chinese (Traditional) | Czech | English (United States) | French | German | Italian | Japanese | Korean | Polish | Portuguese (Brazil) | Russian | Spanish | Turkish". The page is in English. The version is 17.11.1. The page includes a "Download Microsoft ODBC Driver 17 for SQL Server (x64)" link and a "Download Microsoft ODBC Driver 17 for SQL Server (x86)" link. The release number is 17.11.1.1, released on April 30, 2026. The page also mentions that the driver can be installed in the following languages:

- Release number: 17.11.1.1
- Released: April 30, 2026

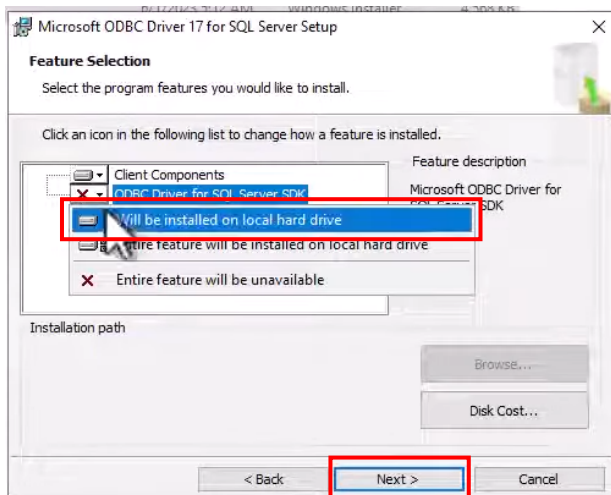
Click next



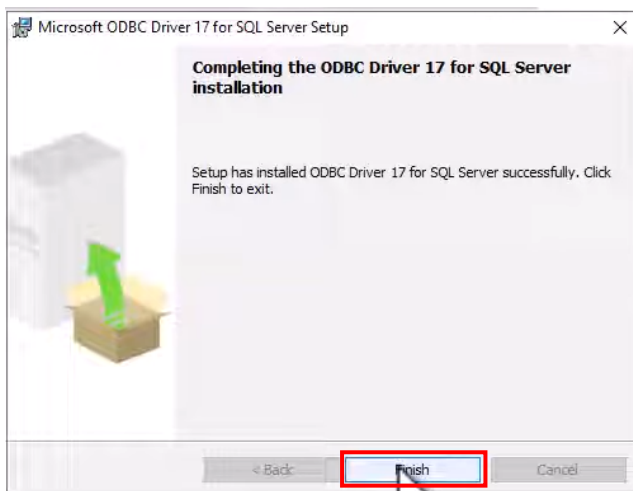
Check: I accept the term in the license agreement > Next



Click ODBC Driver for SQL Server SDK > Will be installed on local hard drive > Next

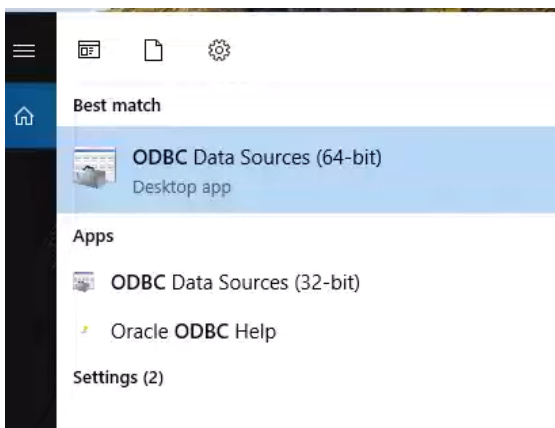


Next > Finish

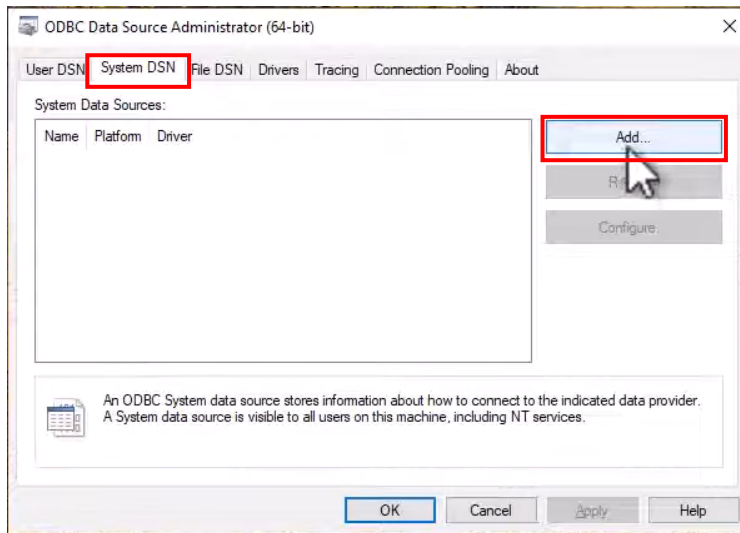


Configure ODBC

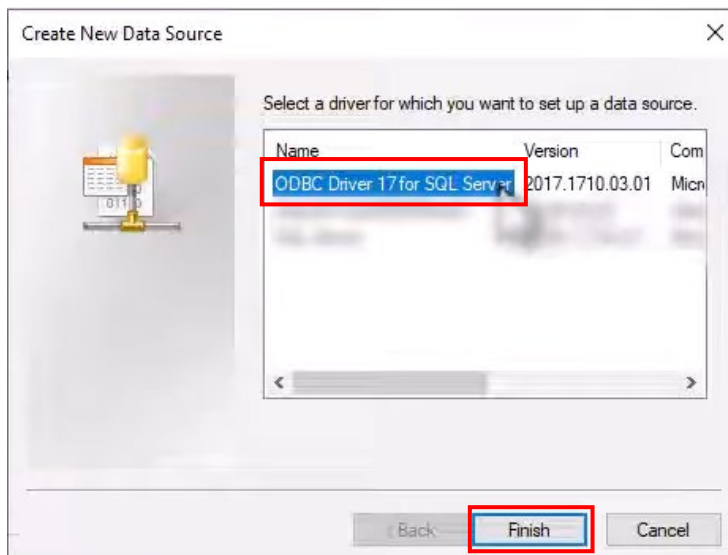
7. Search ODBC Data Sources (64-bit)



Click System DSN > Add



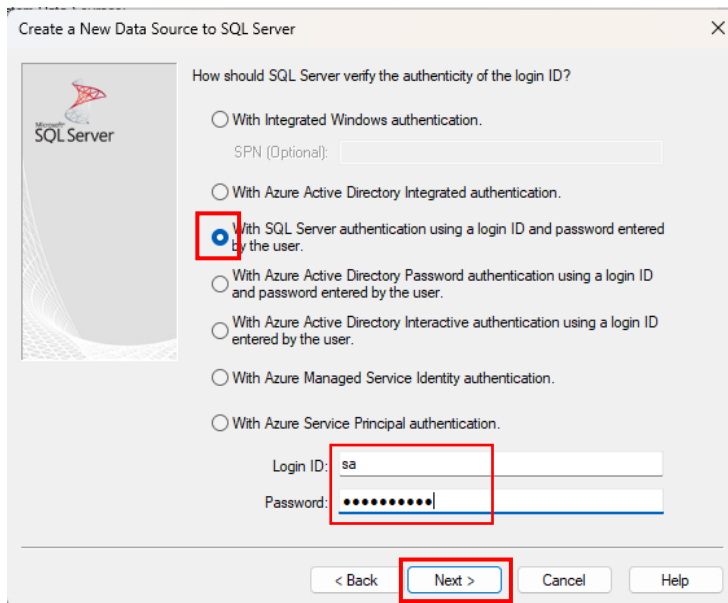
Choose ODBC Driver 17 for SQL Server



Fill in the details accordingly

Name: NRT_MONITOR
Description: NRT Monitoring DB
Server: <Choose your ssms server name>
> Next

If your server is authenticated checked: “With SQL Server authentication using a login ID and password entered by the user”
Enter your login ID and password > Next



Create a New Data Source to SQL Server

How should SQL Server verify the authenticity of the login ID?

- ☐ With Integrated Windows authentication.
SPN (Optional):
- ☐ With Azure Active Directory Integrated authentication.
- ☒ With SQL Server authentication using a login ID and password entered by the user.
- ☐ With Azure Active Directory Password authentication using a login ID and password entered by the user.
- ☐ With Azure Active Directory Interactive authentication using a login ID entered by the user.
- ☐ With Azure Managed Service Identity authentication.
- ☐ With Azure Service Principal authentication.

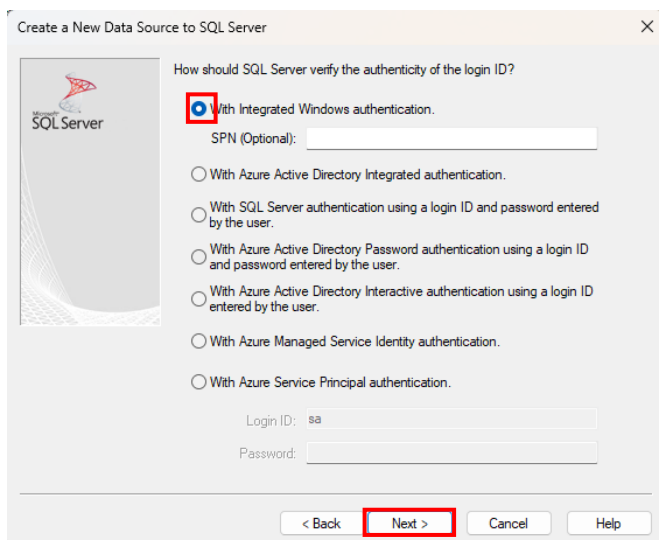
Login ID: sa

Password:

< Back Next > Cancel Help

Login ID: sa
Password: qwerty7890

If your server is unauthenticated, choose “With Integrated Windows authentication” > Next



Create a New Data Source to SQL Server

How should SQL Server verify the authenticity of the login ID?

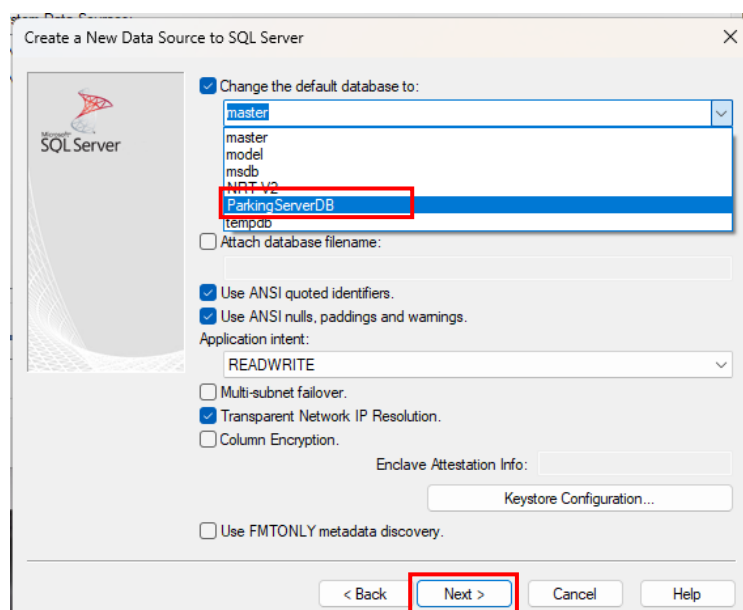
- ☒ With Integrated Windows authentication.
SPN (Optional):
- ☐ With Azure Active Directory Integrated authentication.
- ☐ With SQL Server authentication using a login ID and password entered by the user.
- ☐ With Azure Active Directory Password authentication using a login ID and password entered by the user.
- ☐ With Azure Active Directory Interactive authentication using a login ID entered by the user.
- ☐ With Azure Managed Service Identity authentication.
- ☐ With Azure Service Principal authentication.

Login ID: sa

Password:

< Back Next > Cancel Help

Change the default database to: ParkingServerDB > Next



Create a New Data Source to SQL Server

☒ Change the default database to:

master
master
model
msdb
NTLDR
ParkingServerDB
tempdb

☐ Attach database filename:

☒ Use ANSI quoted identifiers.
☒ Use ANSI nulls, paddings and warnings.

Application intent:
READWRITE

☐ Multi-subnet failover.
☒ Transparent Network IP Resolution.
☐ Column Encryption.

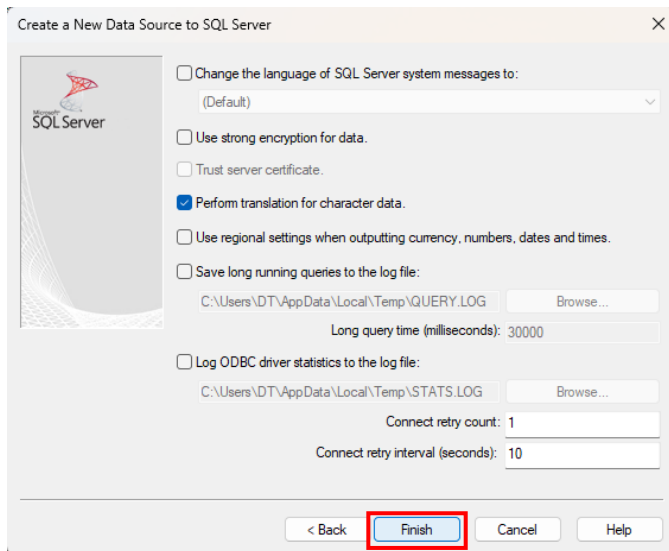
Enclave Attestation Info:

Keystore Configuration...

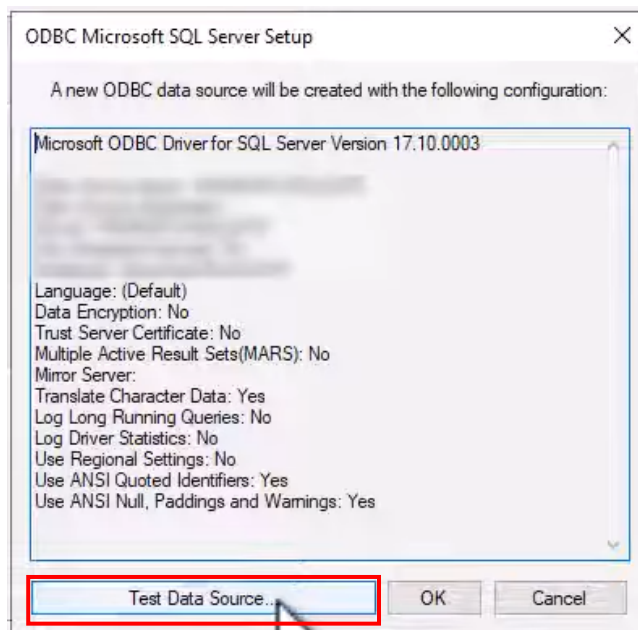
☐ Use FMTONLY metadata discovery.

< Back Next > Cancel Help

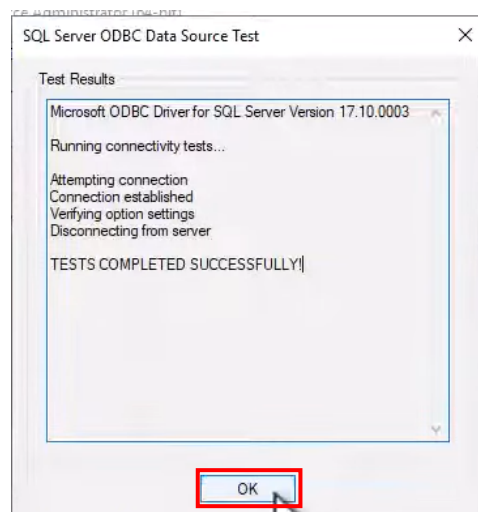
Click Finish



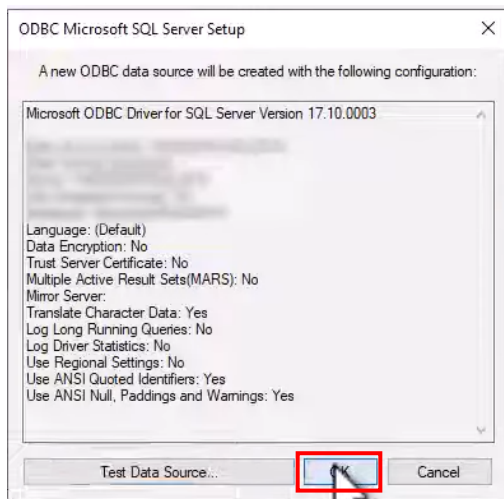
Click Test Data Source



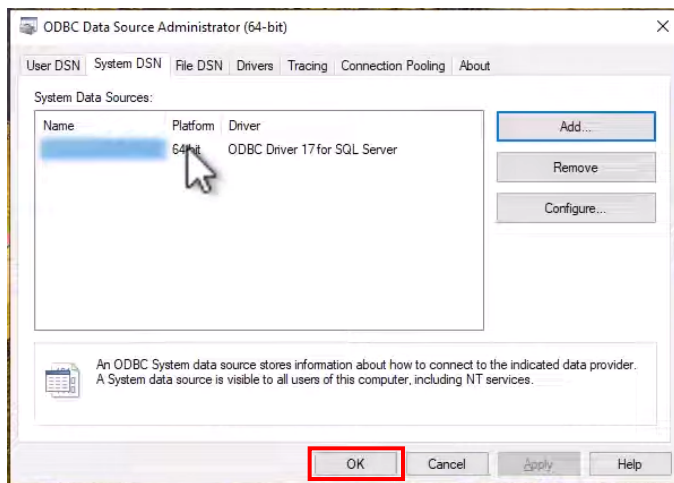
If you see 'Connection established', it means the connection is successful > click OK



Click OK



Click OK



Run Instruction

1. ****Change directory****

```
cd NRT_Monitor
```

2. ****Create a virtual environment****

```
py -m venv venv  
.\venv\Scripts\activate # On Linux Mac OS: source venv/bin/activate
```

- If errors occurred, run this:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

3. ****Install dependencies****

```
pip install -r requirements.txt
```

4. **Credentials configuration**

In the Agent.py

```
SQL_SERVER = r"(local)\SQLEXPRESS" # eg.,: "(local)\SQLEXPRESS" or "DESKTOP-TAICUPD\SQLEXPRESS"  
DATABASE = "NRT-V2"  
SITE_NAME = " eg.,: Penang Courtyard "  
SITE_CODE = DATABASE  
API_URL = "https://script.google.com/macros/s/AKfycbXlAwFbkot1Q5X54EbhLik3NLq85KcpYVkeOyh-  
9Rjgui0nrcI5zlvQaVckHqWAXZju/exec"  
API_KEY = "nrt_8F2xQ9mL7vP3zK1cR6wT4yH0bN5sJ8"  
DASHBOARD_API_URL = "http://127.0.0.1:5000/api/nrt-status"  
  
ALERT_THRESHOLD_HOURS = 48  
COLLECTION_INTERVAL_MINUTES = 40  
  
CONNECTION_STRING = (  
    "DRIVER={ODBC Driver 17 for SQL Server};"  
    f"SERVER={SQL_SERVER};"  
    f"DATABASE={DATABASE};"  
    "UID=sa;"  
    "PWD=qwerty7890;"  
    "Trusted_Connection=yes;"  
)
```

5. ****Run the app****

Place both program one in the Central Server, another in the client site

On the Central Server, run:

```
python app.py
```

On the client side, run:

```
python agent.py
```

Output:

127.0.0.1:5000/?site=test-site-nrt-v2-local-sqlexpress#site-detail

☆

Sites below currently have failed NRT detection based on TB_TNG_CONTROL_BATCH.

Test Site

5 control batch rows

SITE DETAIL

Test Site (NRT-V2)

(local)\SQLEXPRESS | NRT-V2

NRT alert - 5 control batch records pending, oldest operdate is 120 hours old

TERMINAL COUNT

1

BATCH COUNT

5

LATEST OPERDATE

2026-05-07 15:29:46

OLDEST OPERDATE

2026-05-03 15:29:46

OLDEST AGE HOURS

120

RECEIVED AT

2026-05-08 15:48:00

Red

TB_TNG_CONTROL_BATCH

5 rows

Site ID	Terminal ID	Batch No.	Operdate	EOJ Date Time	Record Count	Age Hours	Status
D67	P01	105	2026-05-07 15:29:46	2026-05-08 15:29:46	20	24	
D67	P01	104	2026-05-06 15:29:46	2026-05-07 15:29:46	15	48	
D67	P01	103	2026-05-05 15:29:46	2026-05-06 15:29:46	8	72	
D67	P01	102	2026-05-04 15:29:46	2026-05-05 15:29:46	12	96	
D67	P01	101	2026-05-03 15:29:46	2026-05-04 15:29:46	10	120	